

Bhargav Amin

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Professional Summary

Data Science Master's student (UT Arlington, graduating Dec 2026) specializing in machine learning pipelines, predictive modeling, and AI automation. Recently designed and deployed production ML systems that reduced model retraining time by 40% and improved forecast accuracy by 3%. Seeking roles in data science, ML engineering, or AI where I can own end-to-end pipeline development and drive measurable business impact.

Education

The University of Texas at Arlington / *Masters, Data Science*

Jan 2025 – Dec 2026

P P Savani University / *Bachelor of Technology, Computer Science Engineering (AIML)*

Aug 2020 – May 2024

Work Experience

Enlighten Infosystems / *Data Science Intern*

Dec 2023 – Dec 2024

- Designed 3-model ensemble (TimesFM, Prophet, PatchTST) for demand forecasting on high-volume retail data, improving prediction accuracy by 3% and driving measurable revenue lift.
- Deployed end-to-end automated ML pipeline (GCP Vertex AI, Docker, Airflow) that reduced model retraining from manual weekly cycles to automated daily runs, cutting time-to-production by 40%.
- Built reliable ETL and monitoring infrastructure using Apache NiFi, Airflow, and Superset, enabling real-time dashboards and data-driven decision-making for stakeholders across 3+ business units.
- Published peer-reviewed research paper on customer churn prediction in banking (International Journal of Engineering and Techniques, Vol. 10, Issue 2, 2024) co-authored with company founder.

Bodhivriksh Bizsol LLP / *Web Intern*

Jun 2022 – Jul 2022

- Built full-stack HR management system (PHP/Flask backend, MySQL database) automating employee workflows and reducing manual data entry by ~60%.
- Designed responsive web interfaces and secure API integrations, working directly with HR and IT teams to gather requirements and deliver production-ready features.
- Implemented role-based access controls and real-time reporting functionality enabling instant visibility into employee data across the organization.

Projects

Image Caption Generator using InceptionV3 and Transformer

- Implemented image-to-caption deep learning pipeline using InceptionV3 encoder and Transformer decoder on Flickr8k dataset, generating contextually accurate captions from arbitrary images.
- Achieved competitive BLEU-4 scores on benchmark dataset; demonstrated real-world applicability in document annotation and accessibility use cases.

SandStorm Semantic Segmentation Model using U-Net Architecture

- Built U-Net semantic segmentation model for meteorological storm intensity classification, segmenting satellite imagery into 4 distinct intensity layers with high pixel-level accuracy.
- Deployed trained model for real-time storm impact assessment and geographic risk visualization, enabling rapid meteorological analysis and emergency response planning.

Corrective-RAG – Adaptive RAG with Query Rewriting

- Designed Corrective-RAG system with intelligent query rewriting and adaptive retrieval strategies, achieving 20% higher relevant document recall versus baseline RAG.
- Integrated Chroma vector store, HuggingFace embeddings, and DuckDuckGo fallback search, enabling graceful degradation and improved response quality when retrieval confidence was low.

ML Monitoring System for Production Pipelines

- Built real-time monitoring and observability framework for production ML pipelines, tracking model performance drift, data quality metrics, and system health across distributed environments.
- Integrated alerting and automated retraining triggers, reducing time-to-incident-detection from hours to minutes and enabling proactive model maintenance without manual intervention.

Adversarial Robustness in Vision Transformers

- Investigated adversarial attack vulnerabilities in Vision Transformer (ViT) models and developed defense mechanisms to improve robustness against evasion attacks.
- Achieved 15% improvement in adversarial robustness while maintaining model accuracy, contributing to safer deployments of vision models in critical applications.

Real-time Financial Analytics Dashboard – Coca-Cola Stock Data

- Developed interactive real-time stock analytics dashboard ingesting Coca-Cola market data, implementing automated ETL pipelines for daily updates and multi-timeframe analysis.
- Enabled stakeholders to track market trends, portfolio performance, and risk metrics through customizable visualizations, reducing manual data aggregation time by 80%.

Certifications & Achievements

Google Professional Data Analytics – Coursera

Jan 2023

Published Peer-Reviewed Research: Customer Churn Prediction in Banking – International Journal of Engineering and Techniques

2024

Developed CRM Software for Bodhivriksh Bizsol LLP – Bodhivriksh Bizsol LLP

Summer 2026

Skills

Programming Languages: Python, SQL, C++

Databases: MySQL, ChromaDB

Frameworks & Libraries: TensorFlow, scikit-learn, Pandas, NumPy, Matplotlib, Streamlit

Tools & Software: Docker, Airflow, Apache NiFi, GCP Vertex AI, Tableau, Superset, Power BI

Cloud Platforms: Google Cloud Platform, Microsoft Azure

Soft Skills: ML Pipeline Design, Data Engineering, Cross-functional Collaboration